2D MASWavesPy

This 2D version of MASWavesPy relies mainly on the development of maswavespy a Python package for processing and inverting MASW data, developed by Magnús Snorri Bjarnason at the Faculty of Civil and Environmental Engineering, University of Iceland. The python version depends on four modules: wavefield, dispersion, combination and inversion with two helper modules dataset and select_dc. The 2D version has been extended to deal with the typical CMP setup of a standard 2D MASW survey and provides scripts for CMP sorting, dispersion curve calculation and picking, Bayesian inversion and final model display. Parameter control is provided through a yaml file.

Install the environment using

conda env create -f maswavepy_env.yml

General Usage

Required Input files:

1) Parameter control file (e.g. MASWavespy_config.yaml): Needs definition of site and profile names, the SEG2 inventory with the original shotgather data (in csv, e.g. Line_3000.csv) and the initial velocity model for the Bayesian inversion (e.g. ElCuchillo_initial.csv).
2) SEG2 inventory: Simple text file with path to the original shotgather data, each file on its own line:

```
Example/3006.dat
Example/3007.dat
Example/3008.dat
```

3) A starting velocity file (e.g. ElCuchillo_initial.csv) – NEEDS definition of parameters.

```
layer no,h [m],Vs [m/s],rho [kg/m3],saturated/unsaturated,Vp [m/s],nu [-]
```

```
1,1,150,1850,sat,400,0.3
2,2,175,1900,sat,550,0.3
3,8,200,1950,sat,900,0.3
4 (half-space),,750,1950,sat,1500,0.3
```

The three processing steps can be switched on and off as needed.

```
cmp_sorting:
    enabled: true/false
processing:
    enabled: true/false
plotting:
    create_section: true/false
```

Running the code:

```
python MASWavesPy_processing.py MASWavespy_config.yaml
```

CMP sorting:

CMP spacing is defined as a multiple of the geophone spacing. Eg. CMP bin size twice the geophone spacing

```
cmp_bin_factor: 2
```

CMPs will be stored in output_dir:

```
output_dir: CMP_Gathers
```

For normal CMP processing forward_only should be set to true so that only the forward shots are taken into account to be in line with SeisImager2D.

```
forward_only: true
```

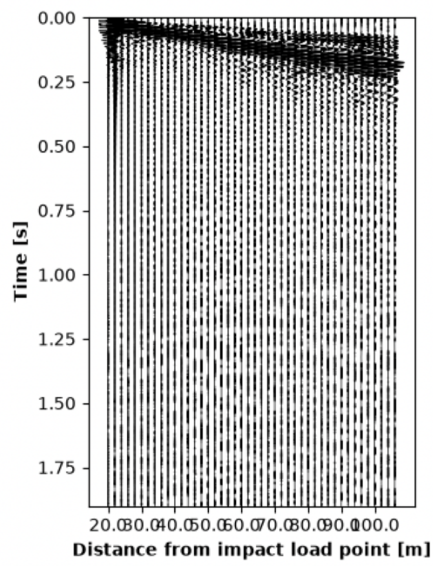
The sorting also generates a geometry file for each CMP which is read when 'cmp' is chosen for reading the dataset.

When skipping the CMP sorting make sure a CMP_inventory.csv is available.

Dispersion curve picking:

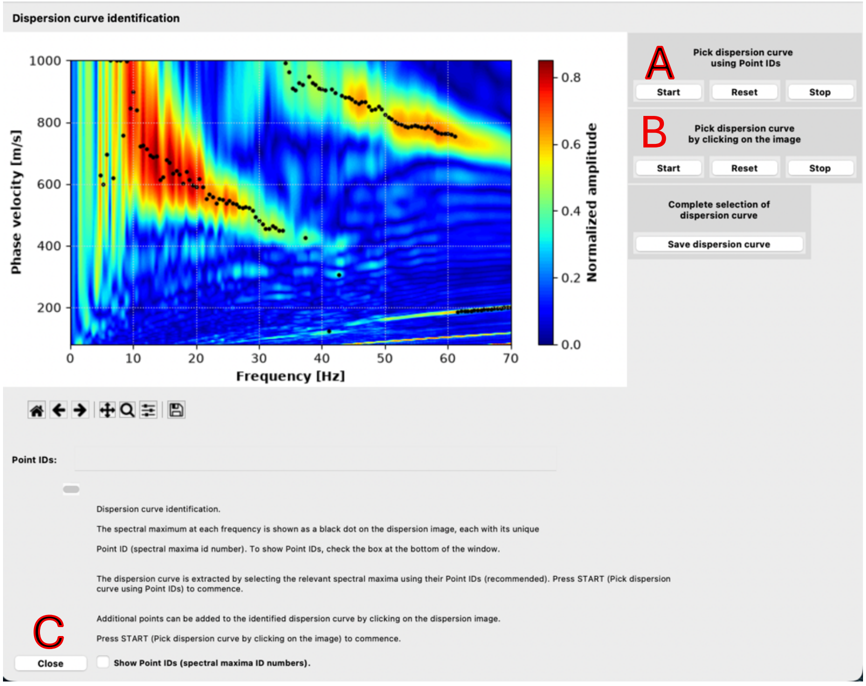
The next step is the picking of the CMP dispersion curves. This needs to be done for each CMP.

1) CMP dataset – The recorded wavefield is show as record section with increasing source receiver distance



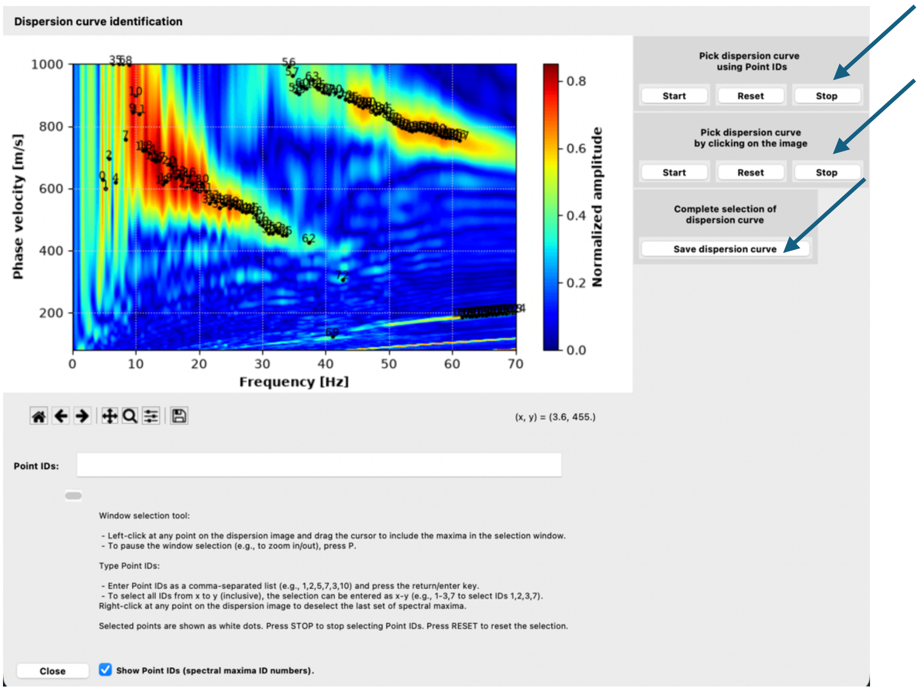
Some information on the CMP (offsets, source/receiver location) is provided on the command lineOnce you have considered the data quality for this CMP close this window through red button on the top left.

2) Now the code calculates the dispersion curve for this CMP and opens up a GUI that allows the manual picking of the dispersion curve.

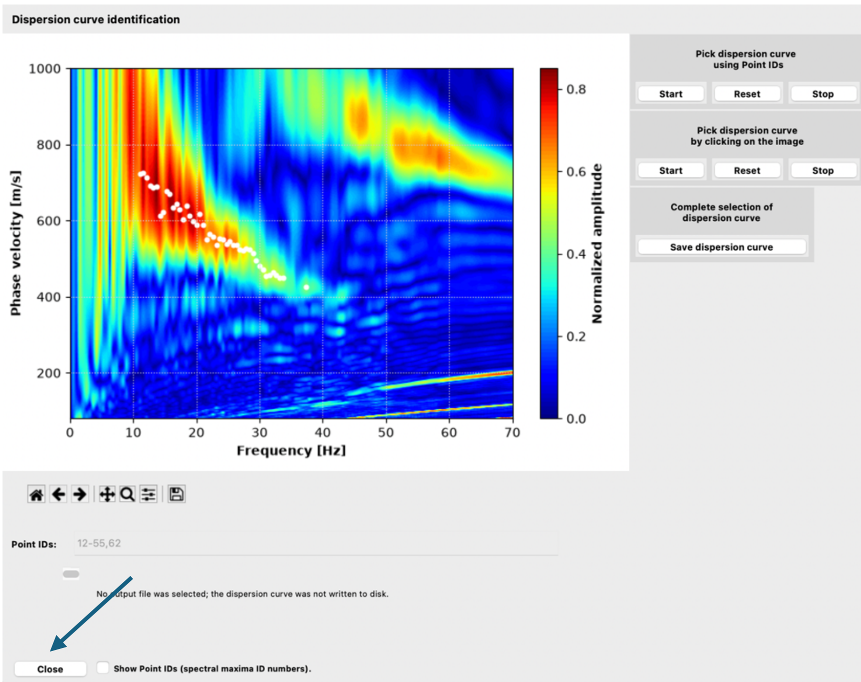


Next, you need to define phase velocities that you want to include into the inversion. Note in this example there is a higher mode (higher phase velocities for frequencies above 40Hz). A) To define a range of frequencies using the mouse through an expandable box click ***Start*** here. B) to define the phase velocities by clicking individual measurements click ***Start*** here. C) Once you clicked on ***Start*** in A or B you can also provide a comma separated list (e.g. 10-15,16,18, …) here.

Once clicked the individual phase velocity measurements get labels and selected frequencies/phase velocity tuples turn white



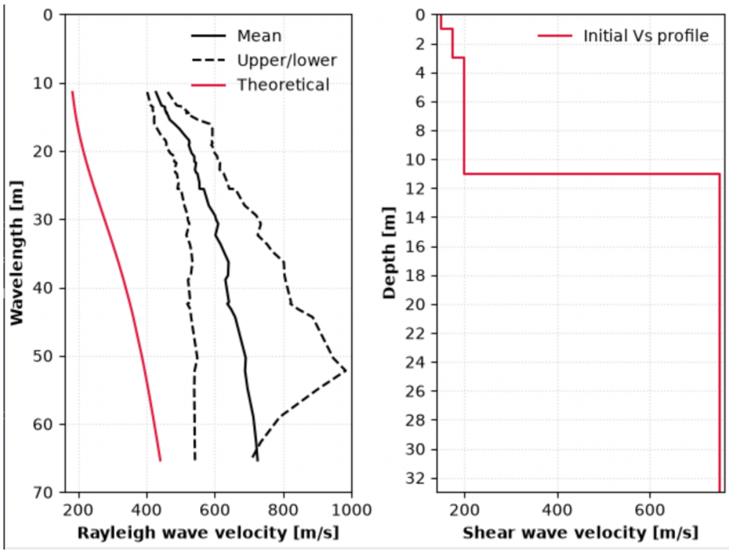
Press ***Stop*** and then ***Save dispersion curve.*** The selected dispersion points are now shown.



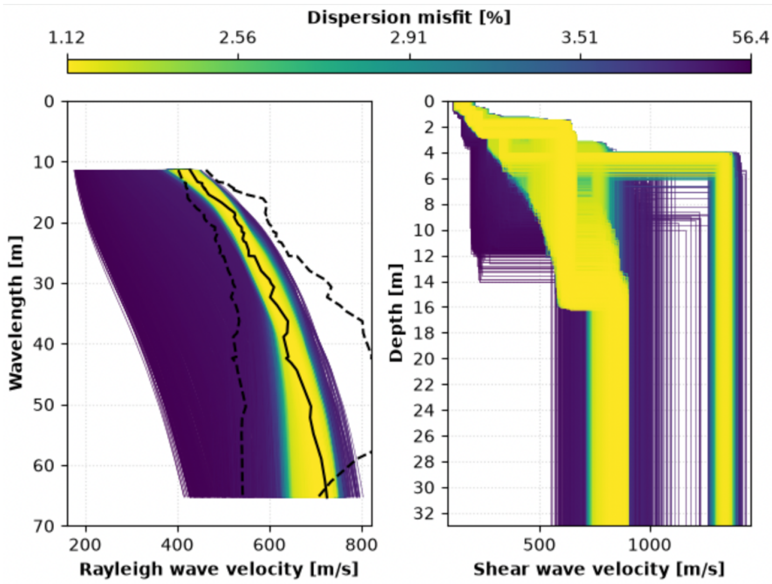
Click ***Close***.

3) Next the picked dispersion curve and the theoretical dispersion curve for the initial velocity model are shown. Not above Point ID 2 was selected with a low phase velocity which leads to the poor fit of the selected and theoretical curve. The initial misfit is shown on the command line. It likely does not matter if the

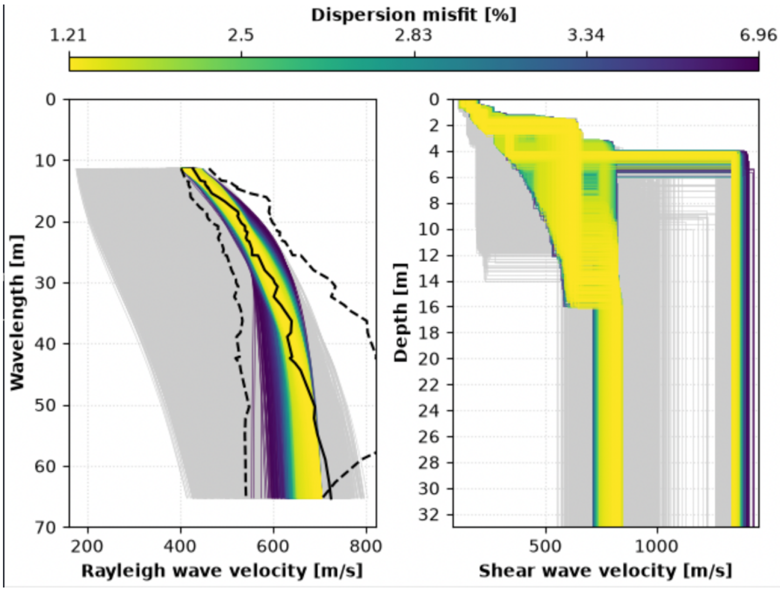
initial fit is poor as long as the search parameters for the inversion are set wide enough.



4) Closing this window will start the runs for the inversion. The window might not close. The command line gives the progress of the inversion
5) Next a couple of windows with the result of the inversion open.
    a. All sampled models that with their misfit. Bright colors are better fitting models with lower misfit. The related Vs velocity structure is shown on the right. Close window after assessing if the sample range is good enough to capture your data.

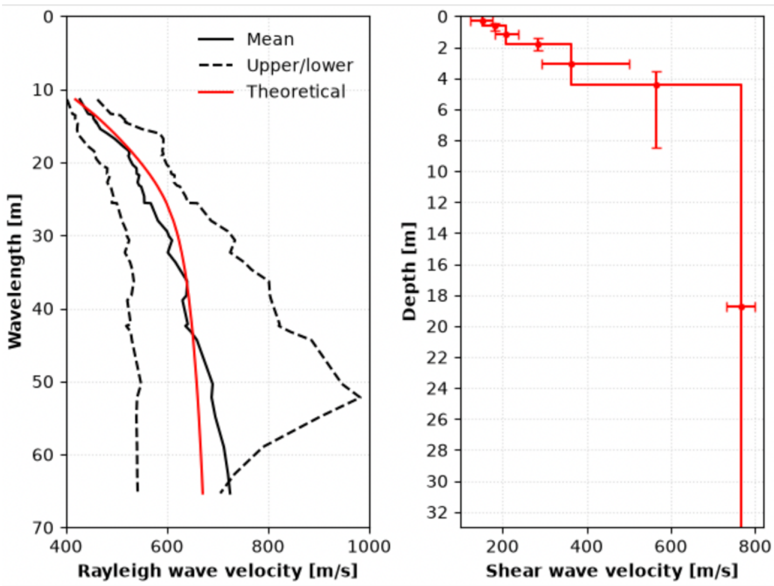


    b. All accepted Vs profile are shown. None selected models are shown in grey. Note in this case the bimodal distribution at depth is likely due to the incorrect pick of the low frequency phase velocity (point ID 2).



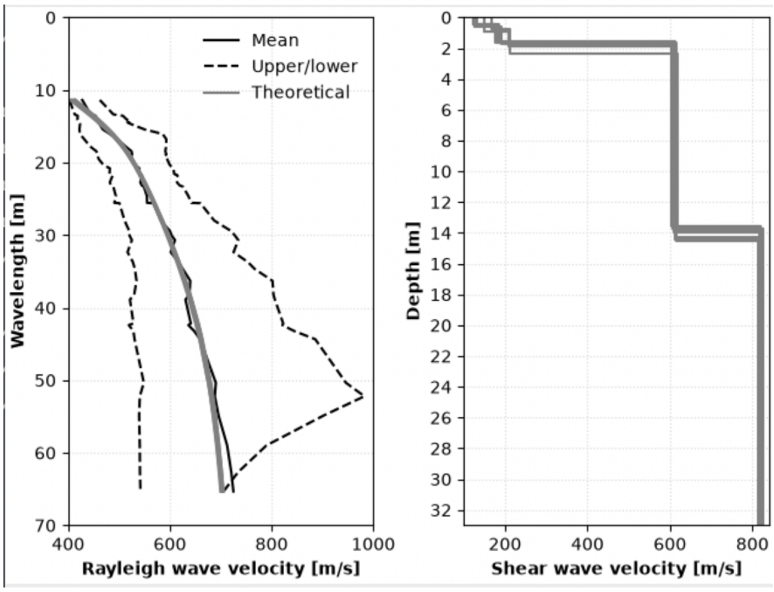
You can close this window, too.

c. Next figure shows the median velocity structure (right) with errors in depth and velocity and the related theoretical dispersion curve (left) for this median model.



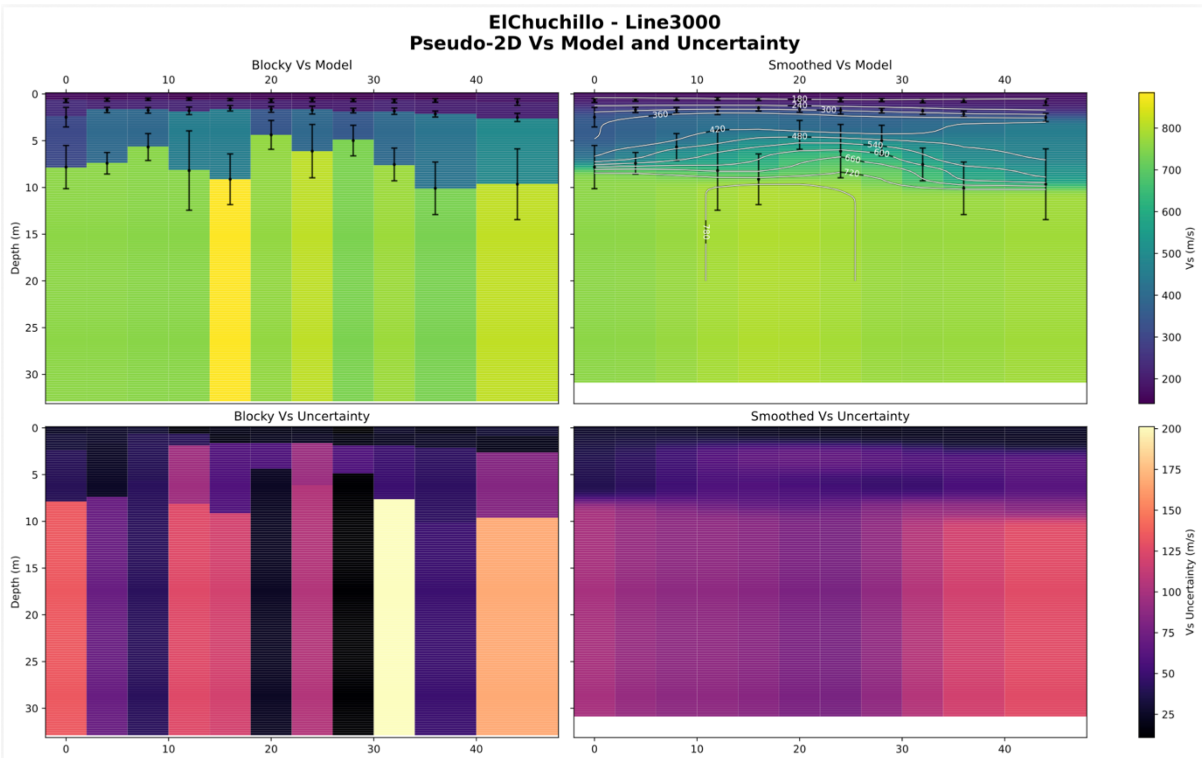
Assess the errors and close the window.

d. Finally, the 10 lowest misfit Vs profiles and the dispersion curves are shown. On the command line some more information about the median velocities at certain depths and the same information for the lowest misfit model is provided to give some idea about the posterior distributions. Close that window (it might not close). The wavefield for the next CMP will automatically be shown.



Plotting the final 2D section

Once all the CMPs have been processed the final 2D section is shown. The two panels are the blocky model showing the way the data actually have been inverted and a smoothed version using a Gaussian filter. The lower panels show the uncertainty in the velocity structure. The ranges in the right panels show the uncertainty in the interface depth from the inversion. All dispersion curves and the modelled structures are saved into the Results directory as pickles.



The final 2D section is also saved as PDF into Results. No other figures are saved and need to be saved, if needed, while you are going along.

Acknowledgements

The backbone of this approach is maswavespy which can be found under this github page: https://github.com/Mazvel/maswavespy

This is based on previous work by:

(1) Park, C.B., Miller, R.D., Xia, J. (1998). Imaging dispersion curves of surface waves on multi-channel record. In *SEG Technical Program Expanded Abstracts 1998*, New Orleans, Louisiana, pp. 1377–1380. https://doi.org/10.1190/1.1820161

(2) Olafsdottir, E.A., Bessason, B., Erlingsson, S. (2018a). Combination of dispersion curves from MASW measurements. *Soil Dynamics and Earthquake Engineering*, 113, pp. 473–487. https://doi.org/10.1016/j.soildyn.2018.05.025

(3) Buchen, P.W., Ben-Hador, R. (1996). Free-mode surface-wave computations. *Geophysical Journal International*, 124(3), pp. 869–887. https://doi.org/10.1111/j.1365-246X.1996.tb05642.x

(4) Olafsdottir, E.A., Erlingsson, S., Bessason, B. (2020). Open-Source MASW Inversion Tool Aimed at Shear Wave Velocity Profiling for Soil Site Explorations, *Geosciences*, 10(8), 322. https://doi.org/10.3390/geosciences10080322

(5) Olafsdottir, E.A., Bessason, B., Erlingsson, S., Kaynia, A.M. (2024). A Tool for Processing and Inversion of MASW Data and a Study of Inter-Session Variability of MASW. Accepted for publication in *Geotechnical Testing Journal,* 47 (5): 1006–1025 10.1520/GTJ20230380